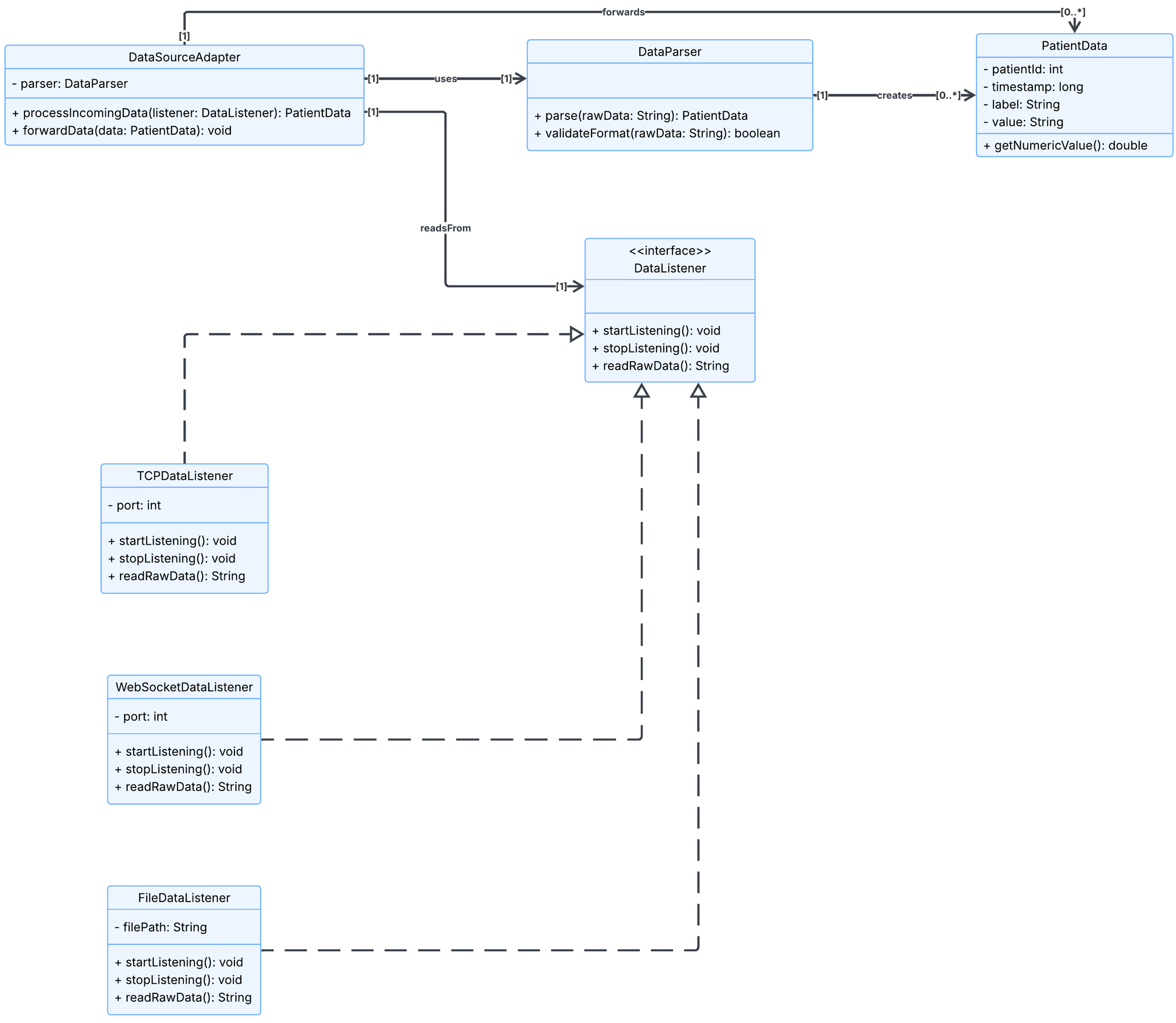
CHMS Data Access Layer – Class Diagram



This diagram models how the CHMS receives external patient data and converts it into a standardized internal format. The design separates transport-specific input, parsing, and adaptation so the rest of the system remains independent of how data arrives.

DataListener is modeled as an interface to support multiple input sources through a common abstraction. TCPDataListener, WebSocketDataListener, and FileDataListener implement this interface, which makes the system easier to extend. For example, a new listener type could be added later without changing the rest of the subsystem. Each listener is responsible only for obtaining raw input from its own source.

DataParser standardizes raw input and converts it into PatientData. This keeps parsing logic separate from transport logic, which improves maintainability and reuse. PatientData represents the normalized record used by the rest of the CHMS, with fields such as patient ID, timestamp, label, and value.

DataSourceAdapter acts as the bridge between listeners and the internal system. It reads raw data through the DataListener abstraction, uses DataParser to transform it, and forwards the resulting PatientData. This design ensures that the rest of the CHMS does not need to know whether data came from TCP, WebSocket, or file input. Overall, the subsystem emphasizes modularity, flexibility, and clean separation of concerns.